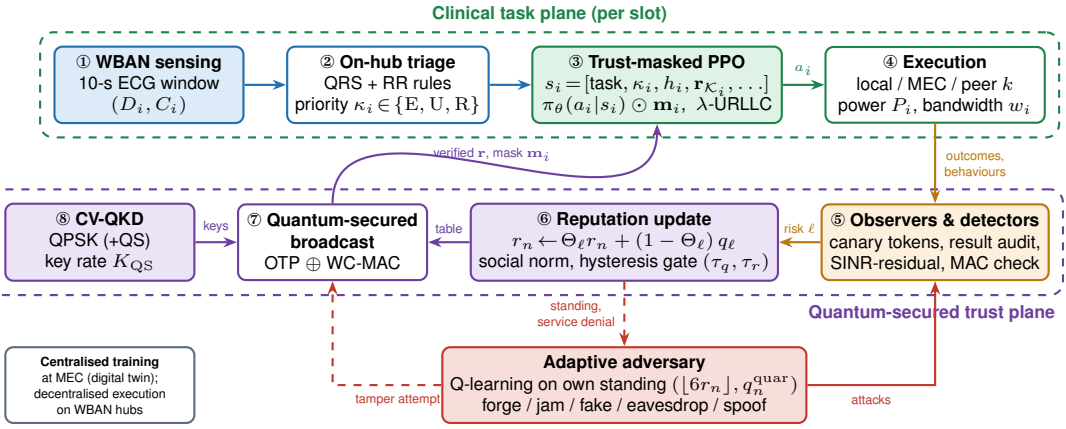

Clinical task plane (per slot)
① WBAN sensing
10-s ECG window
(D_i, C_i)
② On-hub triage
QRS + RR rules
priority $\kappa_i \in \{\mathrm{E}, \mathrm{U}, \mathrm{R}\}$
③ Trust-masked PPO
$s_i = [\text{task}, \kappa_i, h_i, \mathbf{r}_{\mathcal{K}_i}, \ldots]$
$\pi_\theta(a_i|s_i) \odot \mathbf{m}_i$, λ-URLLC
a_i
④ Execution
local / MEC / peer k
power P_i, bandwidth w_i
verified $\mathbf{r}$, mask $\mathbf{m}_i$
outcomes, behaviours
⑧ CV-QKD
QPSK (+QS)
key rate K_{QS}
keys
⑦ Quantum-secured broadcast
OTP ⊕ WC-MAC
table
⑥ Reputation update
$r_n \leftarrow \Theta_\ell r_n + (1-\Theta_\ell)\, q_\ell$
social norm, hysteresis gate (τ_q, τ_r)
risk ℓ
⑤ Observers & detectors
canary tokens, result audit,
SINR-residual, MAC check
Quantum-secured trust plane
standing, service denial
Centralised training
at MEC (digital twin);
decentralised execution
on WBAN hubs
Adaptive adversary
Q-learning on own standing $(\lfloor 6 r_n \rfloor, q_n^{\mathrm{quar}})$
forge / jam / fake / eavesdrop / spoof
tamper attempt
attacks